

Thermo-S1 · Part 1

Progress Tests 1.A → 3.C

Description of a system in thermodynamics

One short test per lecture section, in the format of Test 1

Semester 1 · Academic Year 2026–2027

How to use this booklet

1. Study a section of the lecture notes, then take its progress test *in real conditions*: timer on, no notes, no calculator, answers only inside the frames.
2. Mark yourself with the answer-key booklet. Each circled number in the margin is one marking item; the scheme below each exercise says exactly what earns it.
3. Convert your score: mark out of 20 = score $\times$ 20/14. Write it in the tracker (next page). Below 14/20, re-read the section and retake the test a few days later.
4. When all nine tests are done, take the **End-of-Unit Test** (45 minutes, 50 marks, sections 1.A to 3.C).

Contents

Test	Lecture section	Notes, pages	Duration	Marks
1.A	System and state variables	2–5	7 min	14
1.B	Postulates of thermodynamics	5–7	7 min	14
1.C	Equations of state and state functions	7–10	7 min	14
2.A	Thermoelastic behaviour of condensed phases	11–13	10 min	14
2.B	Thermoelastic behaviour of gases	13–14	7 min	14
2.C	Thermoelastic coefficients: principle and method	14–15	10 min	14
3.A	Low-density gases: the historical laws	15–17	7 min	14
3.B	The ideal gas and Clapeyron's law	17–20	10 min	14
3.C	Equations of state of real gases	20–23	10 min	14

Every test mirrors Test 1 (21 September 2026): a unit conversion, a definition with its equation and SI units, tick-box questions, and a short calculation with the data framed. The data use $R = 8.3 \text{ J K}^{-1} \text{ mol}^{-1}$ and $RT \approx 2.5 \times 10^3 \text{ J mol}^{-1}$ at 300 K, so that everything can be done without a calculator.

Progress tracker

Test	Date	/14	/20	Retake date	/14	To revise
1.A						
1.B						
1.C						
2.A						
2.B						
2.C						
3.A						
3.B						
3.C						
End of unit		/50	/20 = score × 20/50			

Before every test, remember Test 1

- Absolute temperatures in kelvins, $T = \theta + 273.15$, written with a capital K and no degree sign. A temperature *difference* is the same in K and in °C.
- SI units before using $pV = nRT$: 1 L = 10^{-3} m³, 1 cm² = 10^{-4} m², 1 bar = 10^5 Pa, molar masses in kg mol⁻¹.
- Letters first, numbers second, with units; powers of ten separately from the mantissas.
- Before choosing a law, list which variables are fixed and which are free.

Progress Test 1.A — Thermo-S1

1.A · System and state variables

Semester 1 · Academic Year 2026–2027

Family name: Given name:

Duration: 7 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Unit conversion

(1 mark)

Convert the molar volume $V_m = 24.5 \text{ L mol}^{-1}$ into $\text{m}^3 \text{ mol}^{-1}$. Give the exact result, without rounding.

[illegible]

Marking

①	
0	1
☐	☐

Exercise 2. Closed and isolated systems — definitions

(5 marks)

Define a closed system and an isolated system. For a gas enclosed in a cylinder, say what the walls must be like to prevent each kind of exchange with the surroundings. Is every closed system isolated?

Marking

①		
0	1	2

②		
0	1	2

③	
0	1

Exercise 3. Classifying quantities and systems

(4 marks)

Which of these quantities is intensive?

- ☐ the mass m
☐ the volume V
☐ the molar volume $V_m = V/n$
☐ the energy E

A hot-air balloon, with its opening at the bottom, is:

- ☐ closed
 ☐ open
 ☐ isolated
 ☐ closed and isolated

The SI dimension of pressure is:

- ☐ $\text{ML}^{-1} \text{T}^{-2}$
☐ $\text{ML} \text{T}^{-2}$
☐ $\text{ML}^2 \text{T}^{-2}$
☐ $\text{ML}^{-2} \text{T}^{-2}$

 R is an intensive constant. Which relation is homogeneous in the extensive / intensive sense?

- ☐ $\frac{p}{T} = \frac{R}{V}$
☐ $pV = RT$
☐ $\frac{p}{T} = \frac{nR}{V}$
☐ $p = nRT$

Exercise 4. Composition of a solution

(4 marks)

A mass m of sodium chloride (molar mass M) is dissolved in water to obtain a volume V of solution. Compute the amount of substance n of solute, the mass concentration c_m and the molar concentration c . Check that $c_m = cM$.

$m = 5.85 \text{ g}$	$M = 58.5 \text{ g mol}^{-1}$	$V = 0.500 \text{ L}$

Marking

①	
0	1
②	
0	1
③	
0	1
④	
0	1

— End of the test —

Progress Test 1.B — Thermo-S1

1.B · Postulates of thermodynamics

Semester 1 · Academic Year 2026–2027

Family name: Given name:

Duration: 7 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Temperature conversion

(1 mark)

Liquid nitrogen boils at $T = 77\text{ K}$ under atmospheric pressure. Express this temperature in degrees Celsius, exactly.

[illegible]

Marking

1	
0	1

Exercise 2. Thermodynamic equilibrium — postulates

(5 marks)

A system is in contact with surroundings at pressure p_0 and temperature T_0 . Give the two conditions of thermodynamic equilibrium, with the corresponding equalities. State the postulates of classical thermodynamics. When are the state variables of a system defined?

Marking

①		
0	1	2

②		
0	1	2

③	
0	1

Exercise 3. Equilibrium or not?

(4 marks)

A syringe with a free, sealed piston is left for a long time in a room at (p_0, T_0) . The enclosed gas then has:

- | | |
|--|--|
| ☐ $p = p_0$ and $T = T_0$ | ☐ $p > p_0$ and $T = T_0$ |
| ☐ $p = p_0$ and $T > T_0$ | ☐ p and T not defined |

Just after a sudden push on the piston, the pressure of the gas:

- ☐ already has its final value
 - ☐ is equal to p_0
 - ☐ is not defined: it is not uniform
 - ☐ is zero

A metal rod has one end in boiling water and the other in melting ice. Its temperature profile no longer changes. The rod is:

- ☐ at thermodynamic equilibrium
 - ☐ isolated
 - ☐ in a steady state, not at equilibrium
 - ☐ at equilibrium: nothing changes

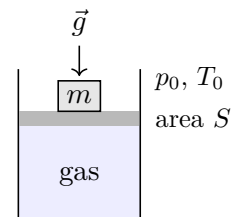
To describe a closed system at equilibrium, classical thermodynamics uses:

- ☐ all positions and velocities
 - ☐ only its mass
 - ☐ a few state variables (p, V, T)
 - ☐ its whole past history

Exercise 4. A loaded piston

(4 marks)

A vertical cylinder is closed by a frictionless piston of area S and negligible mass. The surroundings are at p_0 and T_0 . A mass m is placed on the piston. Find the pressure and the temperature of the gas once equilibrium is reached.



$$p_0 = 1.0 \times 10^5 \text{ Pa} \quad T_0 = 293 \text{ K} \quad m = 1.0 \text{ kg}$$

$$S = 2.0 \text{ cm}^2 \quad g = 10 \text{ m s}^{-2}$$

[illegible]

Marking

1

0	1

2

0	1

3

0	1

4

0	1

— End of the test —

Progress Test 1.C — Thermo-S1

1.C · Equations of state and state functions

Semester 1 · Academic Year 2026–2027

Family name: Given name:

Duration: 7 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Unit conversion

(1 mark)

Convert the pressure $p = 1013 \text{ hPa}$ into bar. Give the exact result.

[illegible]

Marking

①	
0	1
☐	☐

Exercise 2. Equation of state and state function — definitions

(5 marks)

Define an equation of state. Write the total differential dV of the state function $V(T, p)$. Give the meaning of the partial derivative $\left(\frac{\partial V}{\partial p}\right)_T$, and its geometric interpretation in a state diagram.

Marking

①		
0	1	2

②		
0	1	2

③	
0	1

Exercise 3. State diagrams

(4 marks)

The curve $V(p)$ measured at constant temperature is called:

- ☐ an isotherm ☐ an isobar ☐ an isochore ☐ an adiabat

For a low-density gas, $\left(\frac{\partial V}{\partial p}\right)_T$ is:

- ☐ positive ☐ negative ☐ zero ☐ infinite

On the curve $V(T)$ at constant p , $\left(\frac{\partial V}{\partial T}\right)_p$ is:

- ☐ the slope of the curve
 - ☐ the area under the curve
 - ☐ the intercept at $T = 0$
 - ☐ the value of V

An equation of state is valid:

- ☐ during any transformation ☐ only at thermodynamic equilibrium
☐ only for ideal gases ☐ only at 0 °C

Exercise 4. Partial derivatives of a state function

(4 marks)

A gas obeys the equation of state $V = nRT/p$, which is given here. Compute $\left(\frac{\partial V}{\partial T}\right)_p$ and $\left(\frac{\partial V}{\partial p}\right)_T$ in the state below. Then compute the variation dV when T increases by dT and p increases by dp at the same time. Comment.

$$\begin{array}{cccc} n = 1.0 \text{ mol} & R = 8.3 \text{ J K}^{-1} \text{ mol}^{-1} & T = 300 \text{ K} & p = 1.0 \times 10^5 \text{ Pa} \\ & dT = 3.0 \text{ K} & dp = 1.0 \times 10^3 \text{ Pa} & \end{array}$$

Marking

1

01

2

01

3

01

4

01

— End of the test —

Progress Test 2.A — Thermo-S1

2.A · Thermoelastic behaviour of condensed phases
 Semester 1 · Academic Year 2026–2027

Family name: Given name:

Duration: 10 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Unit conversion (1 mark)

The isothermal compressibility of water is $\chi_T = 4.5 \times 10^{-10} \text{ Pa}^{-1}$. Express it in bar^{-1} .

[Empty grid for Exercise 1]

Marking

①
 0 1
 [] []

Exercise 2. Thermoelastic coefficients — definitions (5 marks)

Define the isothermal compressibility χ_T and the isobaric expansion coefficient α of a condensed phase. Give their SI units. Explain the minus sign in χ_T , and why both definitions divide by V .

[Empty grid for Exercise 2]

Marking

①
 0 1 2
 [] [] []
 ②
 0 1 2
 [] [] []
 ③
 0 1
 [] []

Exercise 3. Condensed phases

(3 marks)

For a liquid or a solid, the isothermal compressibility χ_T is:

- ☐ very small ($\chi_T \approx 0$)
 - ☐ negative
 - ☐ very large
 - ☐ infinite

Expressed with the mass density ρ , χ_T equals:

- $$\begin{array}{ll} \square & \frac{1}{\rho} \left(\frac{\partial \rho}{\partial p} \right)_T \\ \square & \rho \left(\frac{\partial \rho}{\partial p} \right)_T \end{array} \qquad \begin{array}{ll} \square & -\frac{1}{\rho} \left(\frac{\partial \rho}{\partial p} \right)_T \\ \square & -\frac{1}{V} \left(\frac{\partial \rho}{\partial p} \right)_T \end{array}$$

Over a restricted temperature range, the coefficient α of a condensed phase can be considered:

- ☐ constant
 - ☐ proportional to p
 - ☐ zero
 - ☐ negative

Exercise 4. Expansion and compression of water

(5 marks)

The coefficients of water are assumed constant. (a) A volume V_0 of water is heated at constant pressure from θ_1 to θ_2 . Estimate the change of volume ΔV . (b) Water is compressed at constant temperature from p_1 to p_2 . Estimate its relative change of volume $\Delta V/V$.

$$\begin{array}{llll} \alpha = 2.0 \times 10^{-4} \text{ K}^{-1} & \chi_T = 5.0 \times 10^{-10} \text{ Pa}^{-1} & V_0 = 250 \text{ mL} & \\ \theta_1 = 20^\circ \text{C} & \theta_2 = 70^\circ \text{C} & p_1 = 1 \text{ bar} & p_2 = 101 \text{ bar} \end{array}$$

Marking

①	
0	1
☐	☐

②	
0	1
☐	☐

③	
0	1
☐	☐

④	
0	1
☐	☐

⑤	
0	1
☐	☐

— End of the test —

Progress Test 2.B — Thermo-S1

2.B · Thermoelastic behaviour of gases

Semester 1 · Academic Year 2026–2027

Family name: Given name:

Duration: 7 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Unit conversion

(1 mark)

A gas has a piezothermic coefficient $\beta = 3.66 \times 10^{-3} \text{ K}^{-1}$. Express β in $^{\circ}\text{C}^{-1}$.

[illegible]

Marking

①	
0	1
☐	☐

Exercise 2. Piezothermic coefficient — definition

(5 marks)

Define the piezothermic coefficient β of a gas. Give its formula, the physical variation it measures, its sign and its SI unit. How does it differ from α ?

Marking

①		
0	1	2

②		
0	1	2

③	
0	1

Exercise 3. Gases at constant volume

(3 marks)

A gas enclosed in a rigid, sealed container is heated. Its pressure:

- ☐ increases ☐ decreases ☐ does not change ☐ becomes zero

The coefficient β measures the variation of:

- ☐ V at fixed p

☐ p at fixed V

☐ T at fixed V

☐ V at fixed T

For a car tyre (airtight, rigid rim), the variables held fixed are:

- ☐ n and V

☐ n and p

☐ T and V

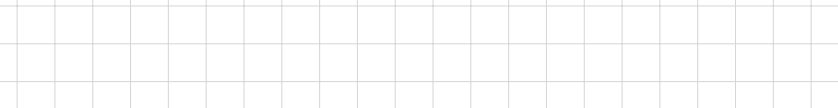
☐ p and V

Exercise 4. Tyre pressure

(5 marks)

A car tyre, whose volume is fixed, is inflated to p_1 at T_1 . While driving, its temperature rises by ΔT . Using the value of β given below (assumed constant), estimate the pressure increase Δp and the final pressure p_2 .

$p_1 = 2.0 \text{ bar}$	$T_1 = 300 \text{ K}$	$\Delta T = 30 \text{ K}$	$\beta = \frac{1}{300} \text{ K}^{-1}$
-------------------------	-----------------------	---------------------------	--



Marking

1

0	1
☐	☐

2

0	1
☐	☐

3

0	1
☐	☐

4

0	1
☐	☐

5

0	1
☐	☐

— End of the test —

Progress Test 2.C — Thermo-S1

2.C · Thermoelastic coefficients: principle and method
 Semester 1 · Academic Year 2026–2027

Family name: Given name:

Duration: 10 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Quick calculation (1 mark)

At $p = 1.0 \times 10^5 \text{ Pa}$, a gas has $\alpha = 3.3 \times 10^{-3} \text{ K}^{-1}$ and $\chi_T = 1.0 \times 10^{-5} \text{ Pa}^{-1}$. Give its piezothermic coefficient β .

[Empty grid for Exercise 1]

Marking

①
 0 1
 [] []

Exercise 2. Relation between the coefficients (5 marks)

State the cyclic relation between the partial derivatives of p , V and T . Use it to show that $\beta = \frac{\alpha}{p \chi_T}$.

[Empty grid for Exercise 2]

Marking

①
 0 1 2
 [] [] []
 ②
 0 1 2
 [] [] []
 ③
 0 1
 [] []

Exercise 3. Method in thermodynamics

(3 marks)

To obtain the equation of state from α and χ_T , one must:

- ☐ differentiate ☐ integrate ☐ multiply them ☐ average them

To obtain α from a known equation of state, one must:

- ☐ differentiate ☐ integrate ☐ invert it ☐ plot it

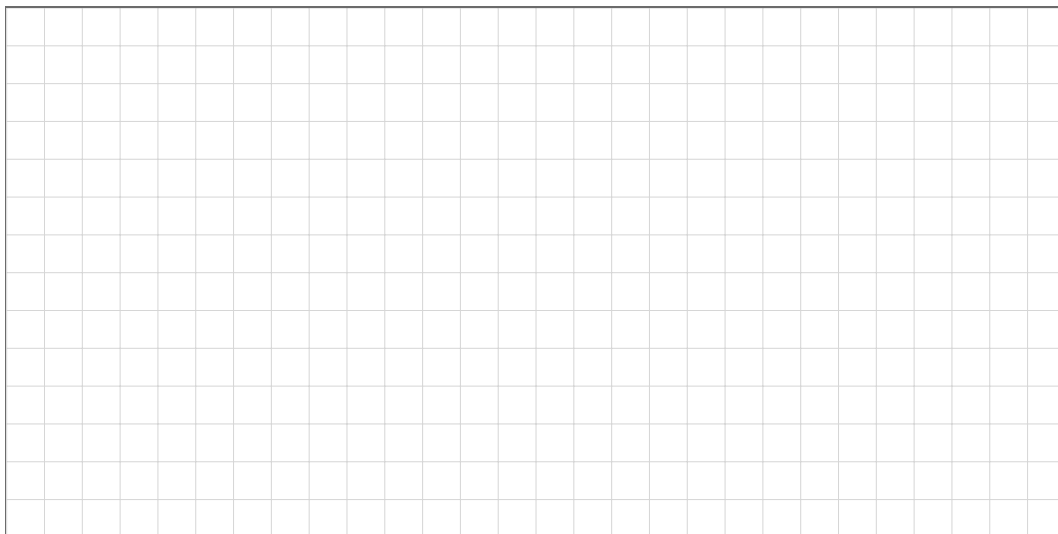
The three coefficients α , χ_T and β are:

- ☐ independent of each other ☐ linked by $\beta = \alpha/(p\chi_T)$
☐ always equal ☐ all dimensionless

Exercise 4. From an equation of state to the coefficients

(5 marks)

A gas obeys the equation of state $p(V - nb) = nRT$, where the co-volume b is a positive constant. Compute α , χ_T and β , then check that $\beta = \alpha/(p\chi_T)$. What do you find when $b \rightarrow 0$?

**Marking**

①
0 1
☐ ☐
②
0 1
☐ ☐
③
0 1
☐ ☐
④
0 1
☐ ☐
⑤
0 1
☐ ☐

— End of the test —

Progress Test 3.A — Thermo-S1

3.A · Low-density gases: the historical laws

Semester 1 · Academic Year 2026–2027

Family name: Given name:

Duration: 7 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Unit conversion

(1 mark)

The product pV of a gas equals 2.0 bar L. Express it in joules.

[illegible]

Marking

①	
0	1
☐	☐

Exercise 2. Boyle–Mariotte law

(5 marks)

State the Boyle–Mariotte law: the conditions under which it holds, and its formula. Deduce from it the isothermal compressibility χ_T of a low-density gas.

Marking

①		
0	1	2

②		
0	1	2

③	
0	1

Progress Test 3.B — Thermo-S1

3.B · The ideal gas and Clapeyron's law

Semester 1 · Academic Year 2026–2027

Family name: Given name:

Duration: 10 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Units

(1 mark)

Express the unit of the gas constant R , $\text{J K}^{-1} \text{mol}^{-1}$, in SI base units.

[illegible]

Marking

1	
0	1

Exercise 2. The ideal gas — definition and Clapeyron's law

(5 marks)

Define the ideal gas model. Write Clapeyron's law, giving the SI unit of every quantity and the value of R .

Marking

①		
0	1	2

②		
0	1	2

③	
0	1

Exercise 3. Controlled and free variables

(3 marks)

For a hot-air balloon (opening at the bottom, rigid envelope), the variables held fixed are:

- ☐ n and V
☐ p and V
☐ n and T
☐ n and p

A weather balloon (sealed, flexible) rises; the outside pressure decreases while its temperature stays about the same. Its volume:

- ☐ increases
 ☐ decreases
 ☐ stays the same
 ☐ becomes zero

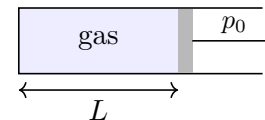
1.0 mol of gas at 300 K and 1.0×10^6 Pa occupies 2.5 L. With $RT \approx 2.5 \times 10^3 \text{ J mol}^{-1}$, its compressibility factor $Z = pV/(nRT)$ is about:

- ☐ 0.1
 ☐ 1.0
 ☐ 10
 ☐ 2.5

Exercise 4. Piston and cylinder

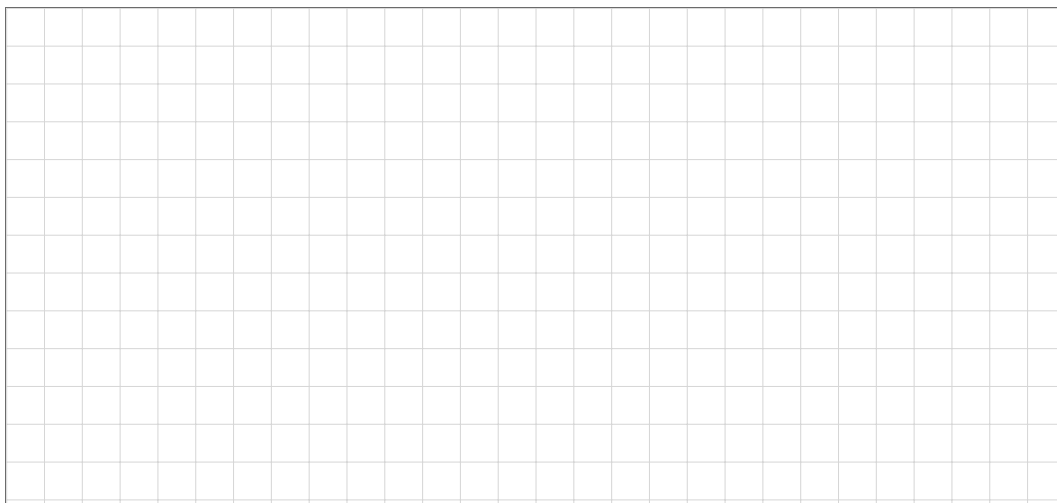
(5 marks)

A horizontal cylinder of cross-section S contains n moles of air, treated as an ideal gas. The piston is initially at L_1 and the temperature is T_1 .



(1) Compute p_1 . (2) The gas is compressed isothermally to L_2 : compute p_2 . (3) The piston is locked at L_2 and the gas is heated to T_3 : compute p_3 . (4) The piston is released against the atmosphere p_0 , at $T_4 = T_3$: compute V_4 and L_4 .

$S = 50 \text{ cm}^2$	$n = 0.10 \text{ mol}$	$L_1 = 50 \text{ cm}$	$L_2 = 25 \text{ cm}$	$T_1 = 300 \text{ K}$
$T_3 = 450 \text{ K}$	$p_0 = 1.0 \times 10^5 \text{ Pa}$	$RT \approx 2.5 \times 10^3 \text{ J mol}^{-1} \text{ at } 300 \text{ K}$		



Marking

①	
0	1
②	
0	1
③	
0	1
④	
0	1
⑤	
0	1

— End of the test —

Progress Test 3.C — Thermo-S1

3.C · Equations of state of real gases

Semester 1 · Academic Year 2026–2027

Family name: Given name:

Duration: 10 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Unit conversion

(1 mark)

The co-volume of carbon dioxide is $b = 4.27 \times 10^{-5} \text{ m}^3 \text{ mol}^{-1}$. Express it in $\text{cm}^3 \text{ mol}^{-1}$.

[illegible]

Exercise 2. The Van der Waals model

(5 marks)

Write the Van der Waals equation of state for n moles. Explain the physical meaning of the term involving b and of the term involving a . What does the equation reduce to when $a \rightarrow 0$ and $b \rightarrow 0$?

Marking

①	
0	1
☐	☐

Marking

①		
0	1	2

②		
0	1	2

③	
0	1

Exercise 3. The Amagat diagram

(4 marks)

In the Amagat diagram, when $p \rightarrow 0$ the compressibility factor Z tends to:

- ☐ 1 ☐ 0 ☐ $+\infty$ ☐ -1

At moderate pressure, $Z < 1$ for most gases because:

- ☐ attractive interactions dominate
 - ☐ the molecules' size dominates
 - ☐ the gas is ideal
 - ☐ the temperature is too high

The Boyle temperature T_B is the temperature at which:

- ☐ $Z \approx 1$ over a wide range of p
 - ☐ the gas liquefies
 - ☐ $Z = 0$
 - ☐ $a = 0$

According to Tab. 7, the gas with the strongest intermolecular attractions, and the highest critical temperature, is:

- ☐
- He
- ☐
- H
- ₂
- ☐
- N
- ₂
- ☐
- H
- ₂
- O

Exercise 4. Carbon dioxide under pressure

(4 marks)

One mole of CO_2 occupies the volume V at temperature T . Compute (a) the pressure p_{id} predicted by the ideal gas law; (b) the pressure p predicted by the Van der Waals equation; (c) the corresponding compressibility factor Z . Interpret.

$$\begin{array}{llll} n = 1.0 \text{ mol} & V = 0.50 \text{ L} & T = 300 \text{ K} & RT \approx 2.5 \times 10^3 \text{ J mol}^{-1} \\ a = 0.36 \text{ Pa m}^6 \text{ mol}^{-2} & b = 4.0 \times 10^{-5} \text{ m}^3 \text{ mol}^{-1} & \text{(rounded values for CO}_2\text{)} & \end{array}$$

Marking

①	
0	1

②	
0	1

③	
0	1

④	
0	1

— End of the test —